

# **Statistical Analysis of Barnsley's Fern : An Application of linear algebra in fractals.**

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# Acknowledgement

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We also thank the Department of Mathematics for providing the opportunity to explore the connection between Linear Algebra, Statistics, and Fractals through this project.



# Abstract

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This project explores the Barnsley's Fern, a fractal generated through Iterated Function Systems (IFS) using four linear transformations with assigned probabilities.

The fern pattern was created using Python and analyzed statistically through mean, variance, standard deviation, and a Chi-square test.

Results show that the random process aligns with theoretical probabilities, confirming the mathematical consistency of the model.

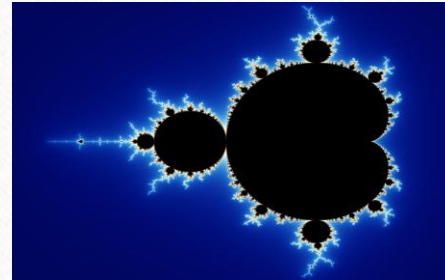
The study highlights how Linear Algebra and Statistics together can explain complex natural patterns through simple mathematical rules.

# Introduction

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## What Are Fractals?

Fractals are complex geometric figures that display self-similarity at every scale. They are created through repetitive mathematical transformations and are often used to model natural patterns like trees, ferns, and clouds.



One of the most famous examples of fractals is **Barnsley's Fern**, which uses linear algebra transformations to generate a realistic fern-like structure.



# Introduction to Barnsley's Fern :

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- **Barnsley's Fern**, introduced by **Michael Barnsley (1988)**, is created using an **Iterated Function System (IFS)** four linear equations that repeatedly transform points with scaling, rotation, and translation based on probabilities.
- Repeating this process thousands of times forms a fractal resembling a **natural fern**, linking **linear algebra** and **statistics** beautifully..

# Objectives

The objective of this project as follows:-

- Generate Barnsley's Fern using linear transformation & (IFS).
- Apply the statistical tools to analyse the distribution of generated data points.
- To test the randomness of the fern using chi square test.
- To summarize the distribution of fractal point using descriptive statistics.
- Visualize the Barnsley's Fern using python programming and interpret the result through statistical analysis.



# Importance Of Barnsley's Fern in Statistics and Mathematics :

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- 1) Fractal Modelling
- 2) Randomness
- 3) Nature Simulation
- 4) Probability



# Mathematical Representation of Barnsley's Fern :

Barnsley's Fern is generated using four linear transformations ( $T_1, T_2, T_3, T_4$ ). Each transformation moves a point  $(x, y) \rightarrow (x', y')$  on the plane using this general rule:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix}$$

where  $a, b, c, d$  are scaling and rotation coefficients, and  $e, f$  are shifting constants.

We use the  $a, b, c, d, e, f$  are the values given by Barnsley's as follows.

Here's a python program to form a fractal: <http://localhost:8888/notebooks/linear%20project%20.ipynb>

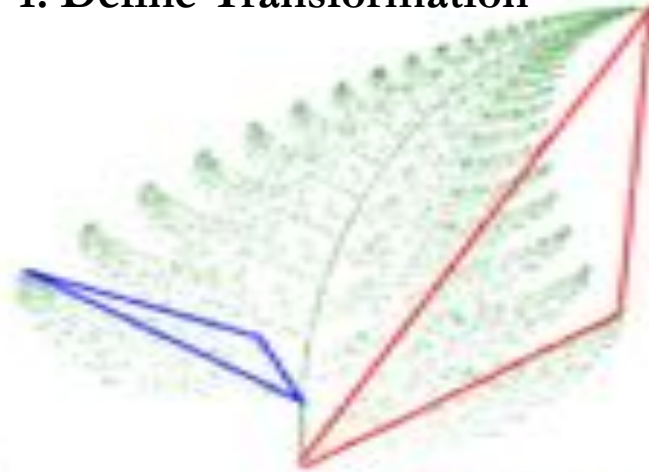
# Transformation Rules :

| Transformation     | Equation                                             | Probability |
|--------------------|------------------------------------------------------|-------------|
| T1(stem)           | $X' = 0x + 0y$<br>$Y' = 0x + 0.16 y$                 | 0.01        |
| T2(Main Leaflets)  | $X' = 0.85x + 0.04y$<br>$Y' = -0.04x + 0.85y + 1.60$ | 0.85        |
| T3(Left Leaflets)  | $X' = 0.20x - 0.26y$<br>$Y' = 0.23x + 0.22y + 1.60$  | 0.07        |
| T4(Right Leaflets) | $X' = -0.15x + 0.28y$<br>$Y' = 0.26x + 0.24y + 0.44$ | 0.07        |



## Fractal fern in four states of construction

1. Define Transformation



2) Apply repeatedly



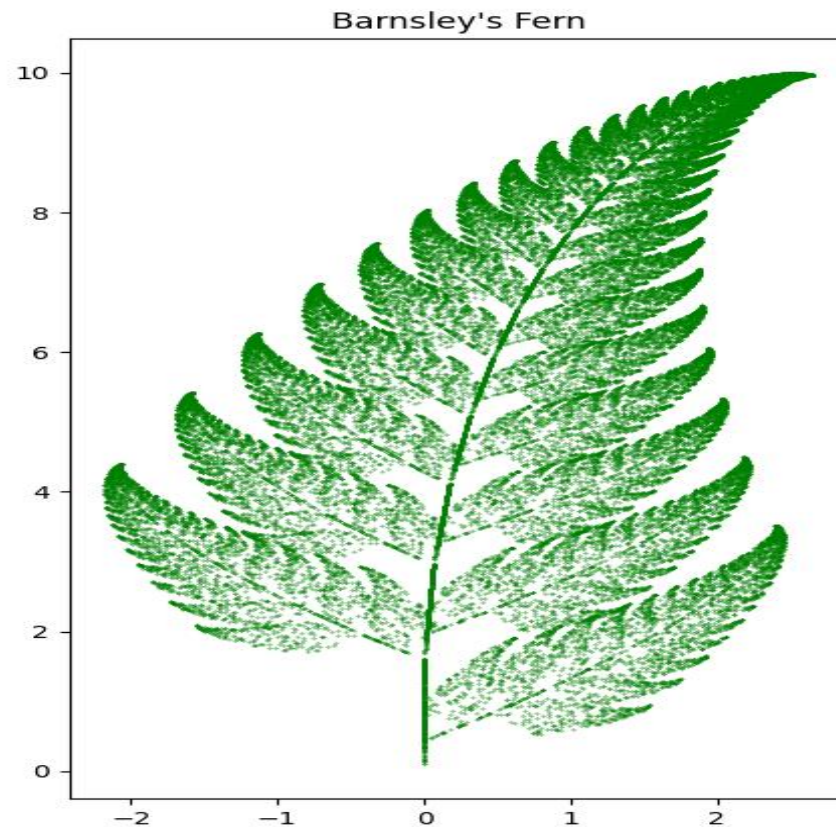
3) Increase Iteration



4) Form Full Fern



# Output Of Barnsley's Fern :





# Statistical Process Behind Fern :

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In this project, statistical concepts were used to analyze and understand the numerical patterns generated by Barnsley's Fern. The process involved the following steps:

1. **Data Generation**
2. **Data Collection**
3. **Descriptive Statistics**
4. **Chi-square Test**
5. **Visualization**

# Descriptive Statistics Values For X & Y

After generating the Barnsley's Fern, the x and y coordinates of all points were analyzed statistically.

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| Measure            | X-values | Y-values |
|--------------------|----------|----------|
| Mean               | 0.8129   | 6.1937   |
| Variance           | 1.4303   | 6.2976   |
| Standard Deviation | 1.19598  | 2.5095   |

The mean and variance show that the fern's x-values are tightly clustered, while the y-values are more spread out indicating the fern's vertical growth pattern.

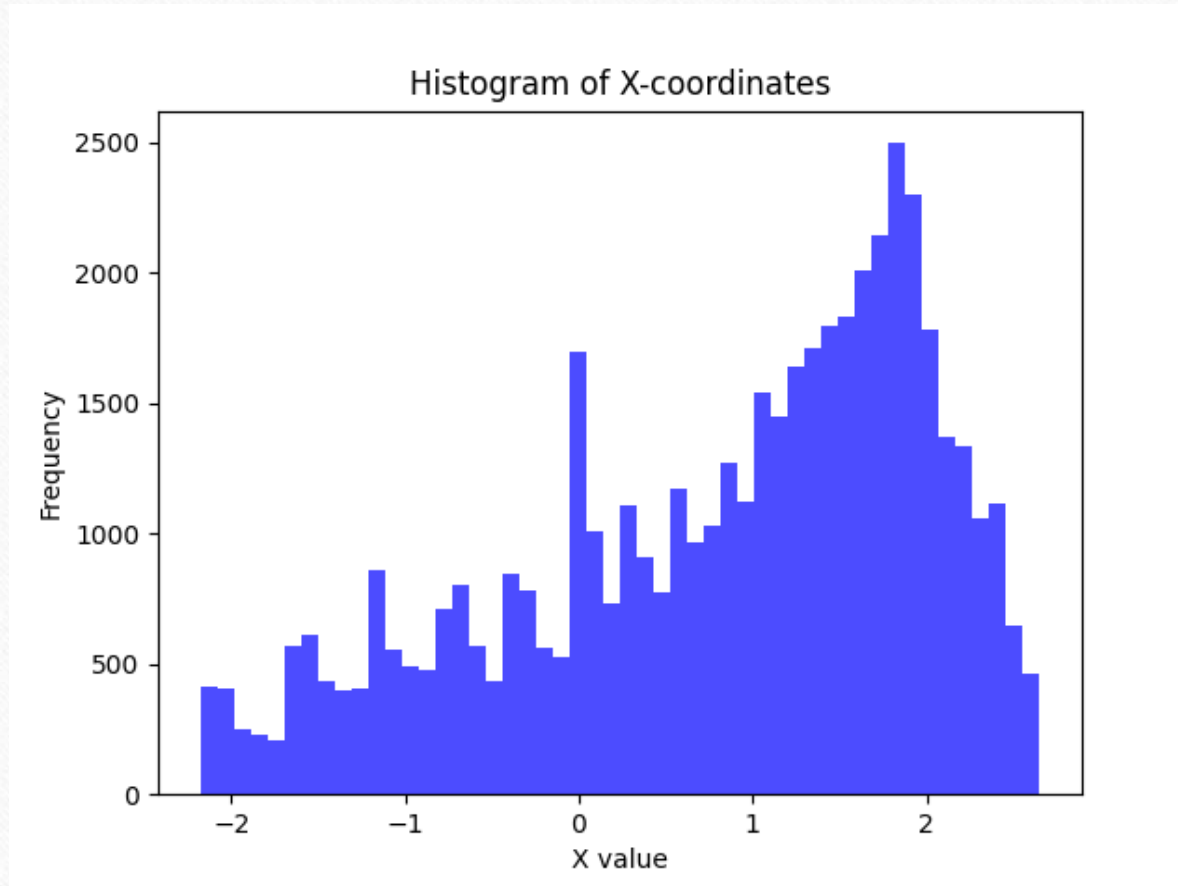


# Chi-Square Test for Transformation Probabilities

| Statistics              | Values                                 |
|-------------------------|----------------------------------------|
| Observed Frequencies:   | [0.01046 , 0.85142 , 0.06852 , 0.0696] |
| Expected probabilities: | [0.01 , 0.85 , 0.07 , 0.07 ]           |
| Chi square Statistics:  | 2.85546                                |
| P-value:                | 0.4144499                              |

**Interpretation:** The chi-square test produce a p-value of 0.4144 which is greater than 0.05 so it is a goodness of fit and accurate fern generate.

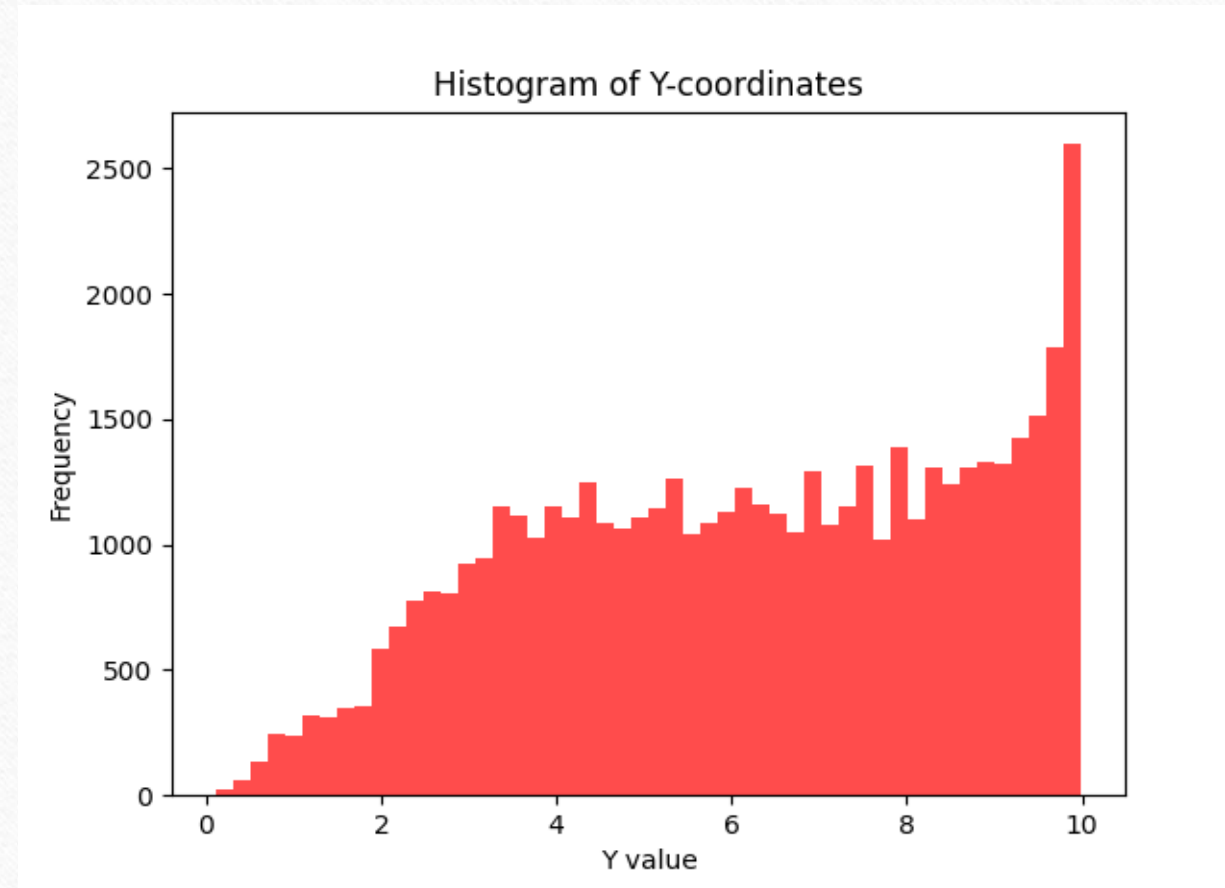
# Histogram of X-Coordinate



This Figure shows that how x-values are symmetrically distributed across the fern's width.



# Histogram of Y-Coordinate



This Figure displays the vertical density pattern, highlighting the fern's height structure.

# Conclusion and Real Life Applications

## Conclusion:-

- ❑ Barnsley's Fern shows how simple math rules create complex natural patterns.
- ❑ Using Iterated Function Systems (IFS) and probability makes realistic fractals.
- ❑ Statistical tests confirm predictable structure with small random variations.
- ❑ The project connects math, statistics, and nature showing order in randomness.



## **Real-Life Applications:-**

- 1. Nature Modelling**
- 2. Computer Graphics & Animation**
- 3. Data Analysis & Statistics**
- 4. Mathematical Education**

# References

- Barnsley, M. F. (1988). Fractals Everywhere. Academic Press.
- Wikipedia – Barnsley Fern  
([https://en.wikipedia.org/wiki/Barnsley\\_fern](https://en.wikipedia.org/wiki/Barnsley_fern))
- NumPy, Matplotlib, and Pandas – Python libraries used for data generation and visualization.
- Statistical formula references – Applied Statistics and Probability for Engineers (Montgomery & Runger).



**Thank you**